

# Arundhati Vasishth

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## EDUCATION

### UNIVERSITY OF MASSACHUSETTS, AMHERST

Amherst, MA

BS, Computer Science

GPA: 3.83

Expected Graduation Date: Fall 2028

**Achievements:** Chancellor's Award, Dean's List Honoree

## SKILLS

**Languages:** Python, Java, TypeScript, JavaScript, HTML/CSS

**Frameworks & Cloud:** React.js, Node.js, Express, Flask, TensorFlow/Keras, AWS (Bedrock, Kendra), MongoDB, REST APIs, Git

**Data & AI:** RAG pipelines, LLM integration, statistical analysis, data modeling, visualization, hypothesis testing

## EXPERIENCE

### Massachusetts Executive Office of Health & Human Services (EOHHS)

Boston, MA

AI Software Engineer

Spring–Summer 2026

- Built an AI-powered regulatory chatbot for Massachusetts DPH's statewide public health platform (Metrik), targeting 85%+ accuracy across 351 local health departments and 5 inspection domains.
- Architected a RAG pipeline on AWS Bedrock with Amazon Kendra, grounding all responses in official Massachusetts CMR documents to eliminate hallucination risk for court-level enforcement decisions.
- Conducted stakeholder interviews with DPH to define KPI benchmarks, reducing inspector document search time from 2–10 minutes per query to near-instantaneous AI responses.
- Designed audit logging, inspector feedback, and flagging systems to continuously monitor response accuracy, completeness, and recall across all inspection domains statewide.

### Indian Institute of Science (IISc)

Bangalore, India.

Machine Learning Research Intern

Summer 2025

- Conducted a comparative study of 6 machine learning and deep learning pipelines (SVM, PCA-SVM, CNN, XGBoost, PCA-XGBoost, and CNN-XGBoost) for medical image classification using 8,700+ chest X-ray and brain MRI images across binary and multi-class tasks.
- Implemented VGG16 transfer learning and hybrid CNN-XGBoost pipelines, achieving up to 96% test accuracy on brain MRI classification and 95% on chest X-ray classification while outperforming standalone SVM and CNN models.
- Evaluated model robustness across multiple train-validation-test splits using accuracy, F1-score, ROC-AUC, and log-loss, and collaborated with 11 researchers to publish findings and identify hybrid deep learning architectures as the most reliable approach for limited-data medical imaging.

### Minutemen Alternative Investment Fund (MAIF)

Amherst, MA

Quant Arm, Junior Analyst

Spring 2026

- Built a multi-agent NLP pipeline semantically matching prediction-market contracts across 4 exchanges (Kalshi, Polymarket, Manifold, Metaculus) to find deterministic arbitrage pairs instead of directional bets.
- Designed a pipeline that embeds market questions into a Qdrant vector DB, clusters via KNN, and uses an LLM (Grok/Llama) to classify semantic relationships across 5 relation types, scoring price mismatches for arbitrage.
- Built a live polling service that re-fetches quotes for confirmed pairs and alerts when profit edge exceeds a configurable threshold.

## PROJECTS

### AI-Powered Hand Gesture Recognition Platform

Amherst, MA

FingerSolve

Spring 2025

- Built a real-time gesture-recognition learning tool for touch-free math quizzes, improving engagement for children and users with disabilities by 40% during pilot testing.
- Engineered a React/Flask platform using MediaPipe Hands + TensorFlow/Keras, achieving 94% gesture-classification accuracy across 12 gestures. Designed a dynamic UI with confidence scoring, reducing user input errors by 30%.
- Deployed a live prototype that earned 2nd place at the MassAI ML Conference and attracted interest from 50+ educators for inclusive learning use cases.

### AI-Powered Medical Results Translator

Amherst, MA

MedEcho

Spring 2026

- Built a 3-portal system where a DenseNet-121 CNN classifies chest X-ray findings and GradCAM highlights the region driving the prediction, turning raw scan results into plain-language explanations in the patient's own language.
- Used Gemini to generate patient scripts and ElevenLabs + HeyGen's LiveAvatar to deliver diagnoses out loud, face-to-face, in 4 languages, both before and after physician review.
- Added a clinical-trial matching feature using the ClinicalTrials.gov API to score patient eligibility, and built the full Flask/React prototype in 24 hours.

### Full-Stack Sustainability Analytics Platform

Amherst, MA

EcoForecast

Spring 2026

- Built a full-stack sustainability platform simulating 20-year financial ROI and carbon impact of green investments for small businesses using custom TypeScript/Node.js financial models.
- Developed backend APIs and MongoDB workflows to process quarterly utility data (electricity, water, fuel) and generate cost projections, break-even analysis, and carbon reduction estimates.
- Integrated a React analytics dashboard with 4+ visualizations and a LLaMA 3.3-powered AI advisor to provide context-aware sustainability insights.